

# Lecture 11 - Loneliness

The Social Brain: Critical Perspectives on Science, Society and Neurodiversity

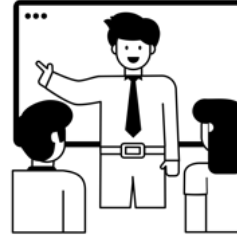
Richard Ramsey



# Today

## Part 1

- Loneliness



## Part 2

- Read articles and discuss



# Overview

- Background
- The reaffiliation model
- Neuroscience
- Brain injury
- Lifespan
- Interventions



# Background



# What is loneliness?

Loneliness and social isolation are often used interchangeably, which is confusing.

**Social isolation is objectively quantifiable:** the quantity and frequency of social contact, plus proximity to others.

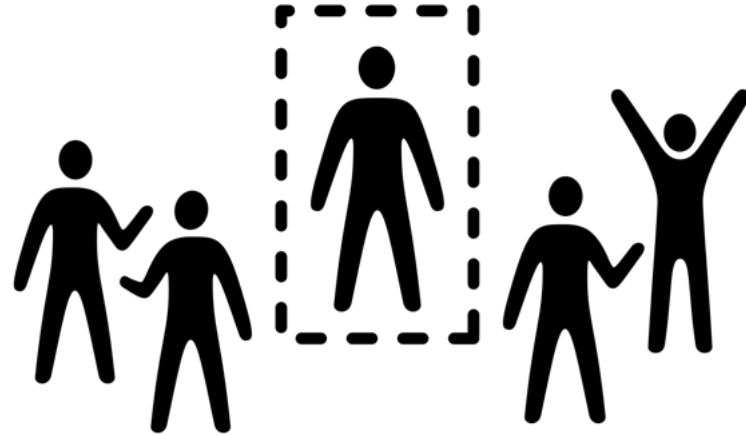
**Loneliness is a subjective interpretation:** an evaluation that social needs are not being met satisfactorily.



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(Cacioppo & Patrick 2008, Perissinotto & Cowinsky 2014, Valtorta et al. 2016)

# Isolation versus loneliness



# Types of loneliness

**Social loneliness:** the absence of a desired social network (family, friends and acquaintances).

**Emotional loneliness:** the lack of attachment figure/s that one could form an intimate connection with.

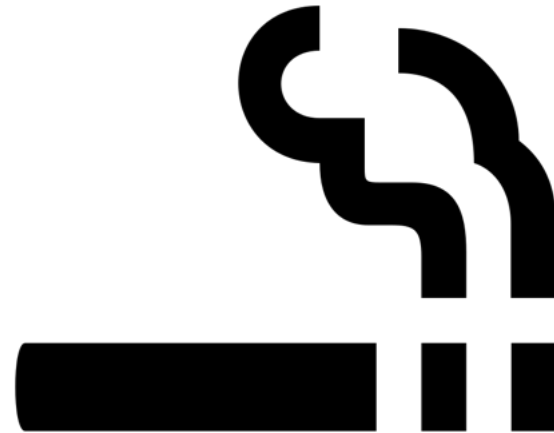


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(Dahlberg & McKee 2014 • Weiss 1973)

# Health impact

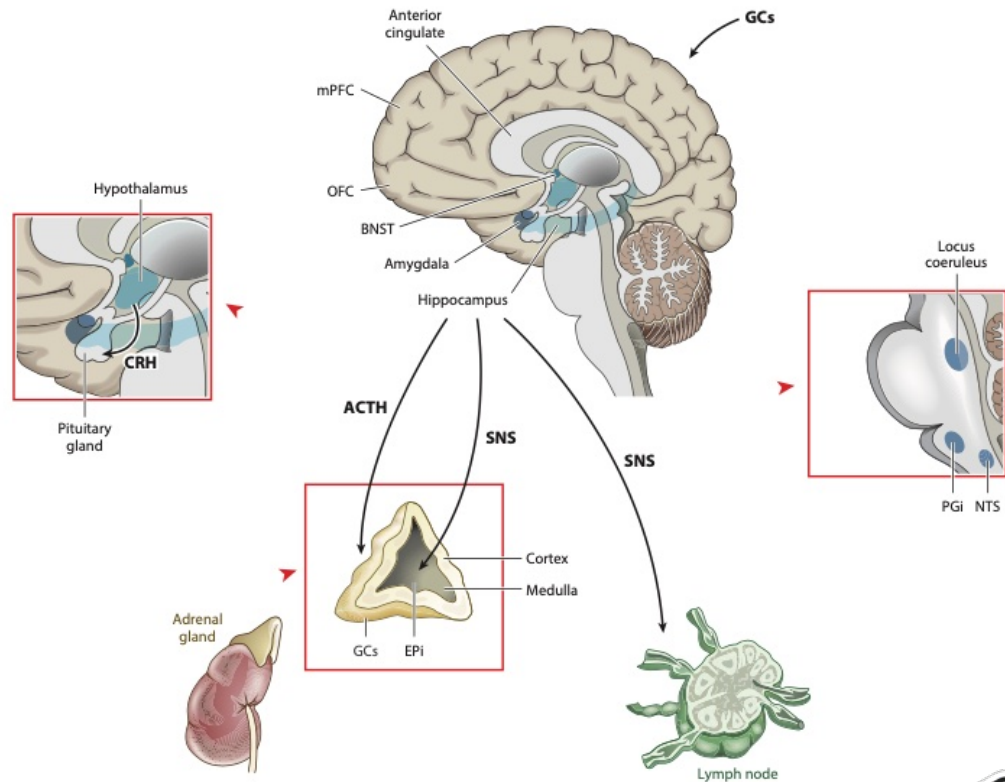
- Loneliness is associated with increased morbidity and mortality.
- It is clearly bad for your health and well-being.
- But is loneliness worse than smoking 15 cigarettes a day?





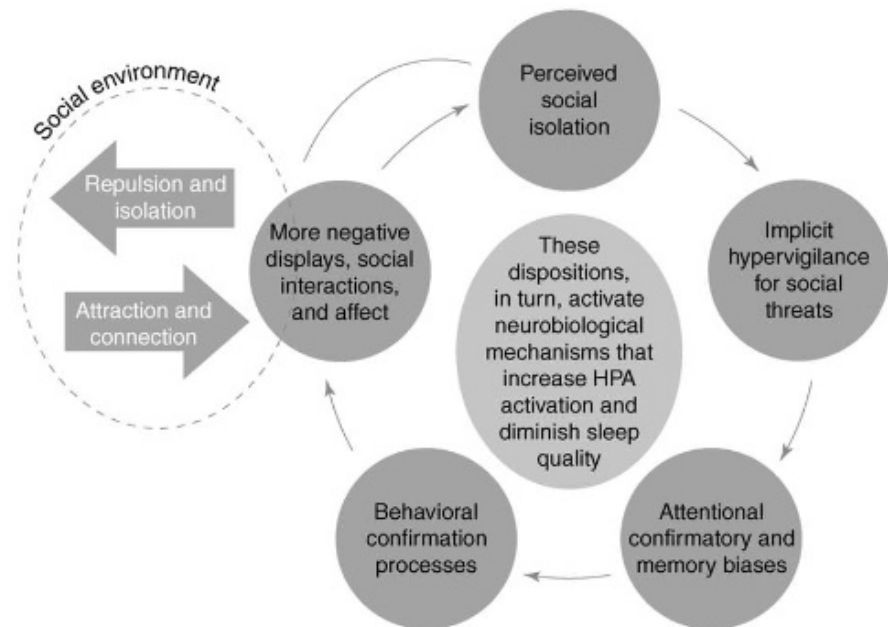
# Physiological impact

Loneliness predicts physiological responses associated with stress (e.g., elevated blood pressure, rise in cortisol), *above and beyond objective measures of social isolation.*



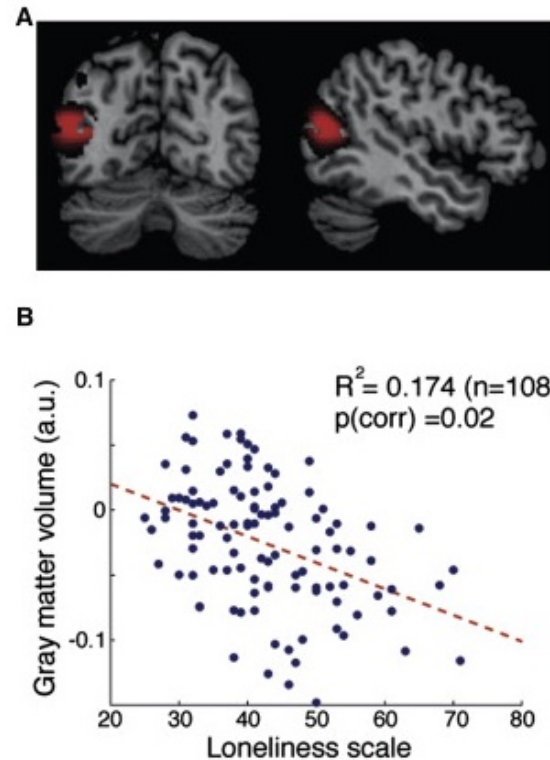
# Psychological impact

Loneliness predicts maladaptive cognition and behavior (e.g., perception of social interactions as unpleasant, reduced physical activity, and lifetime satisfaction), *above and beyond objective measures of social isolation.*



# Basic research impact

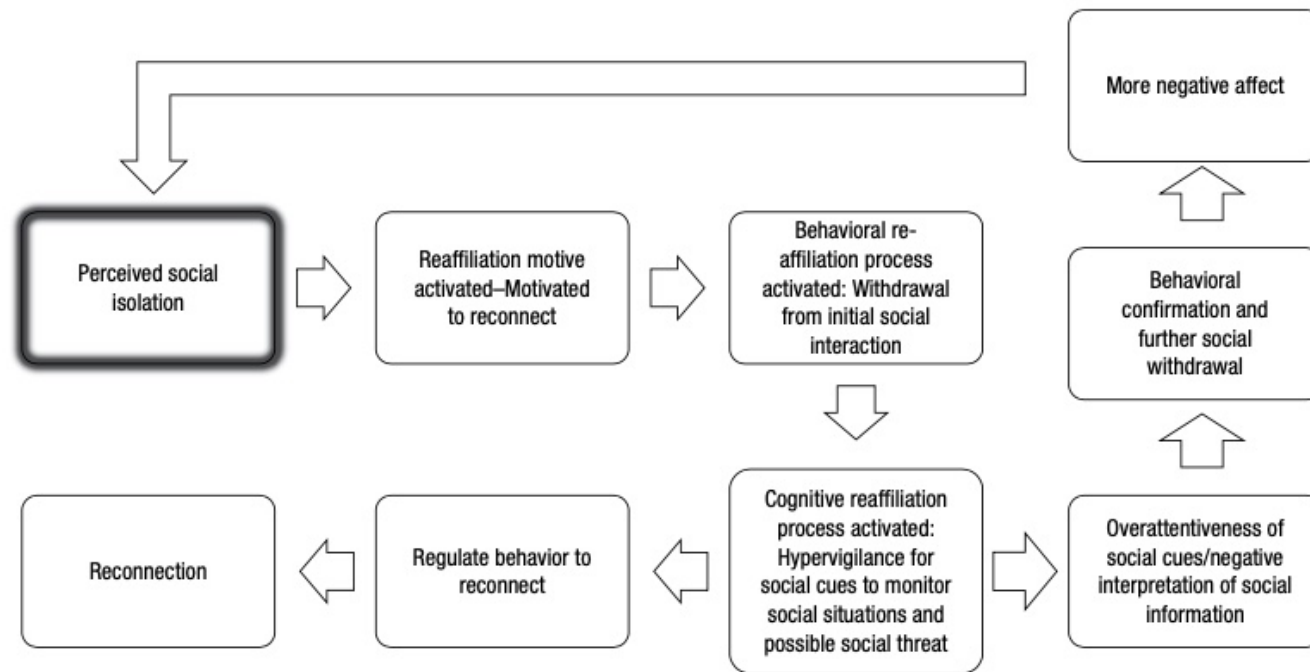
- Loneliness offers a way to study individual differences in social neuroscience.
- Somewhat surprisingly, social neuroscience rarely studies variation in “socialness” across individuals.
- But things are changing...



# The reaffiliation model (RAM)



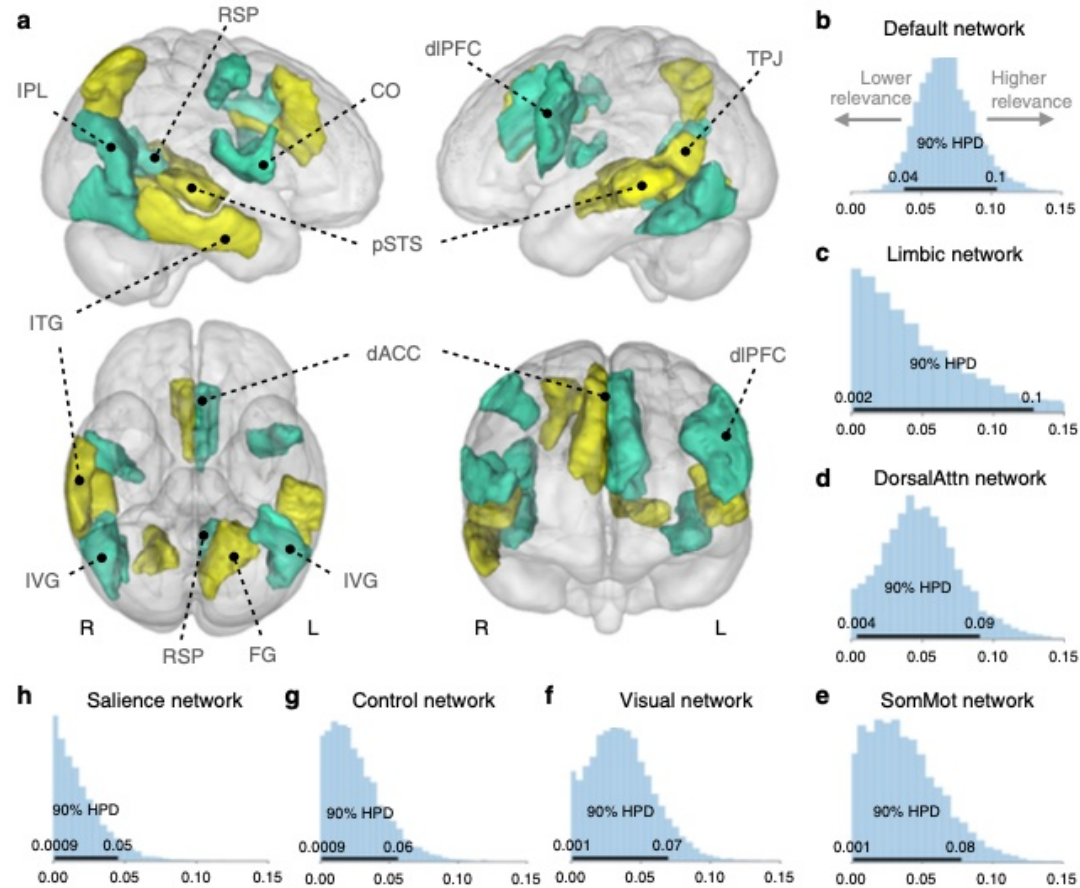
# RAM



# Neuroscience



# Structural changes

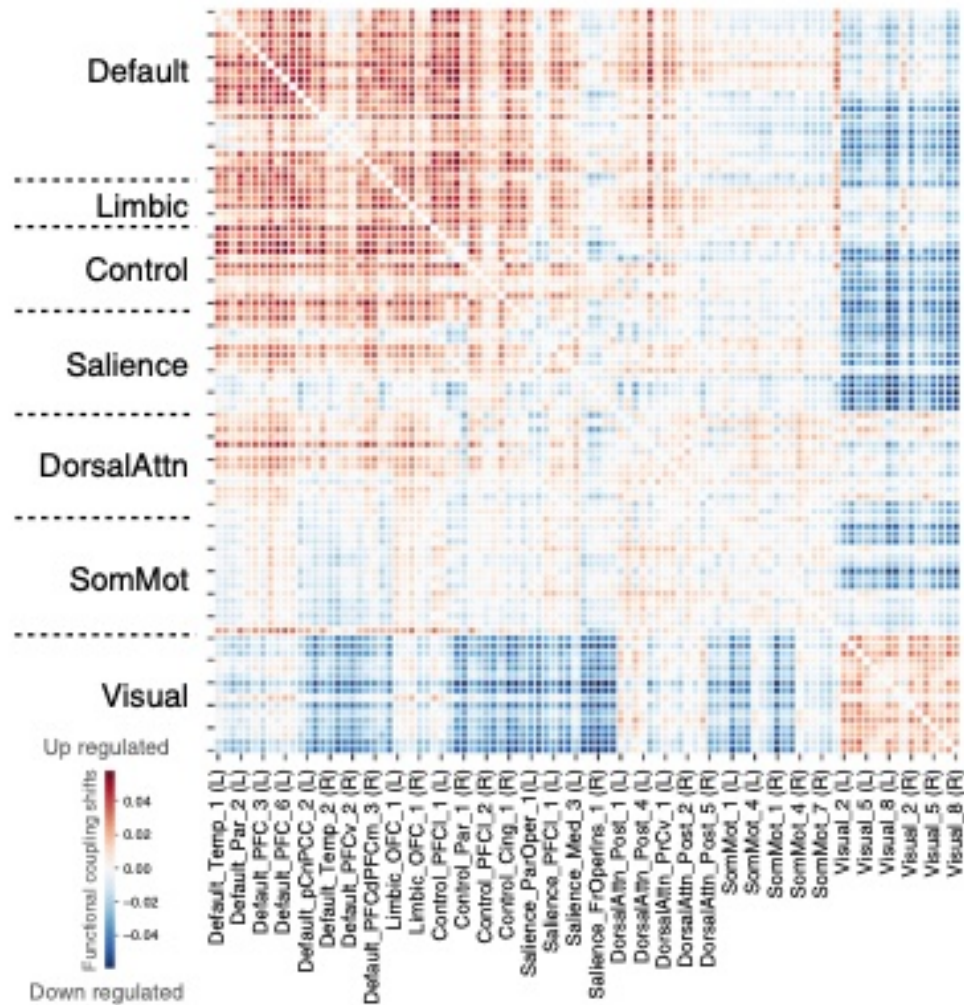


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(Spreng et al. 2020)



# Functional changes



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(Spreng et al. 2020)



# Brain injury



# Clinical observation

- Following a brain injury, such as a stroke, cognitive and social functions are impacted.
- Brain injury assessments routinely include an assessment of cognitive function.
- But what about social function and loneliness?

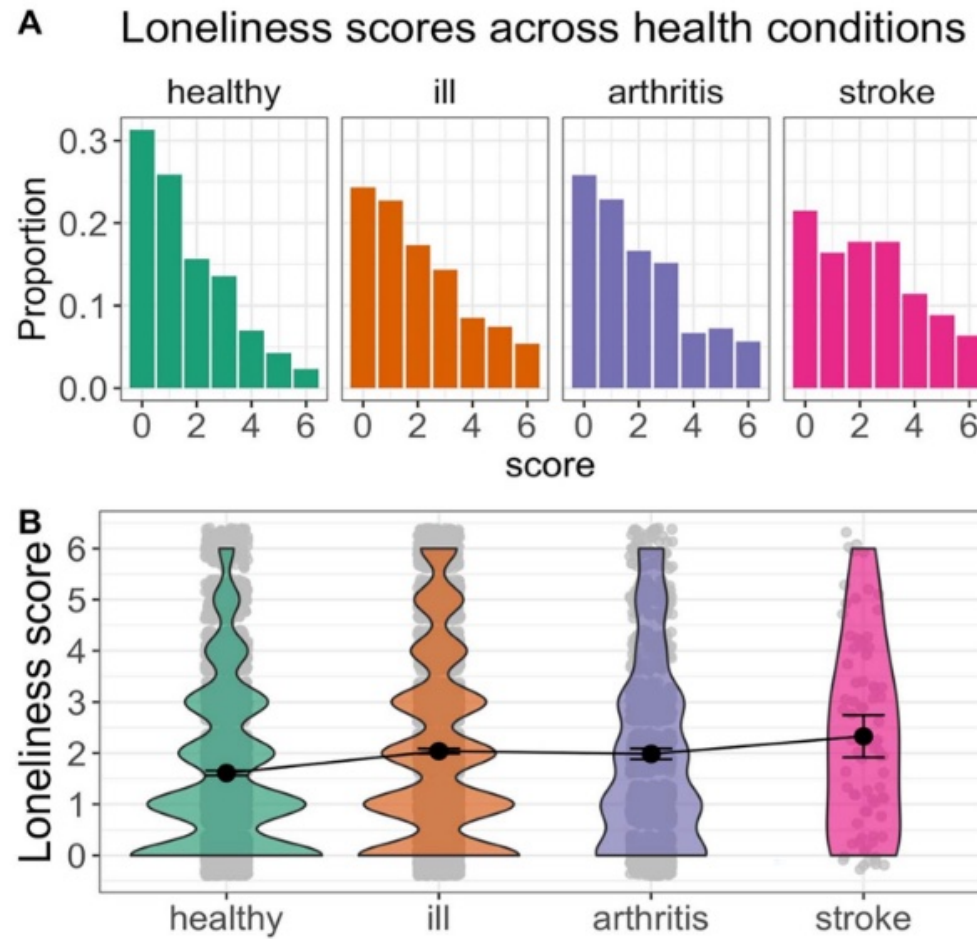


# Assessing loneliness following stroke

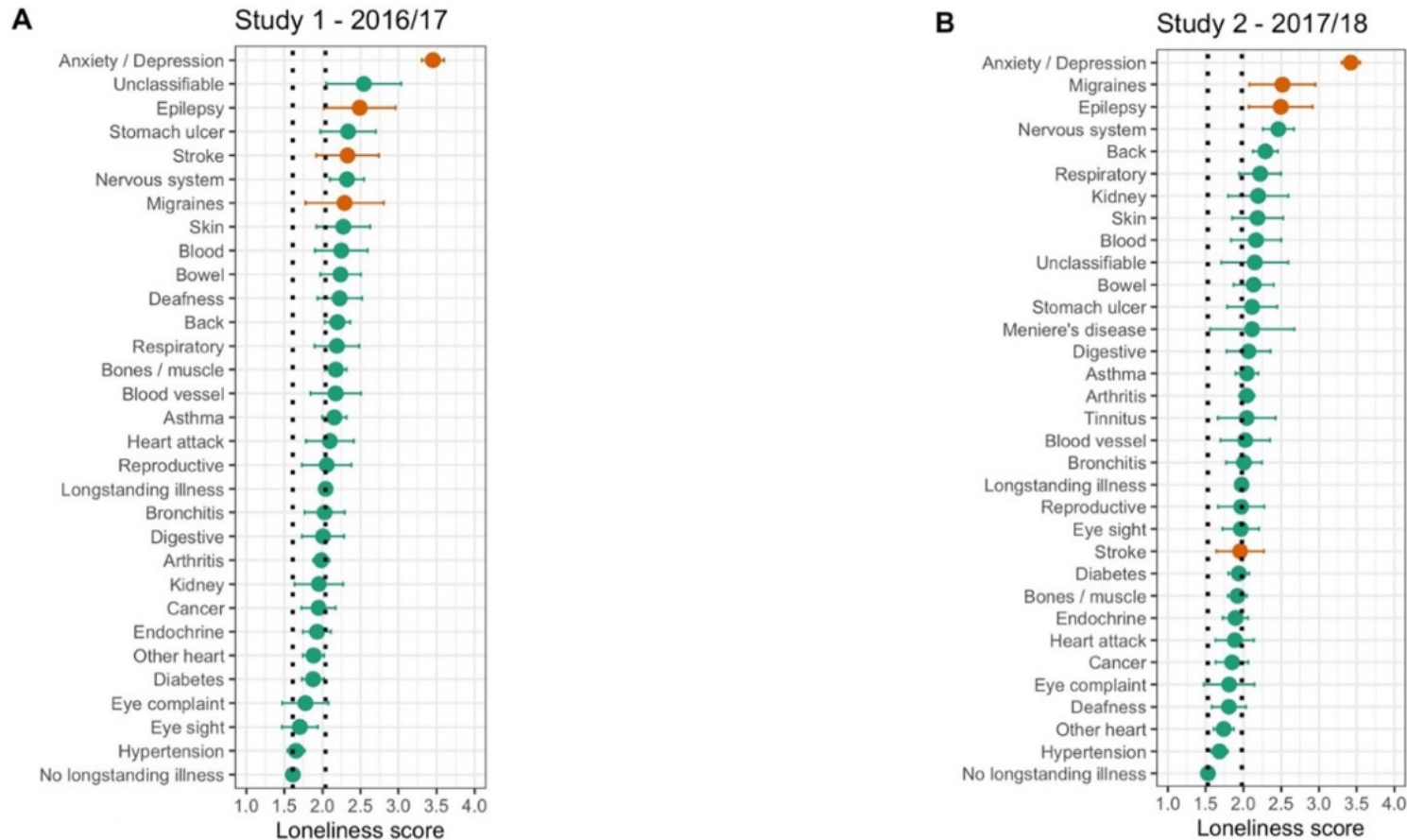
- The national survey for Wales dataset
- Two cohorts (2016/17 and 2017/18)
- De Jong Gierveld Loneliness Scale ([Gierveld & Tilburg, 2006](#))
- N~8000 per cohort.



# Loneliness ratings and stroke



# What about other illnesses?



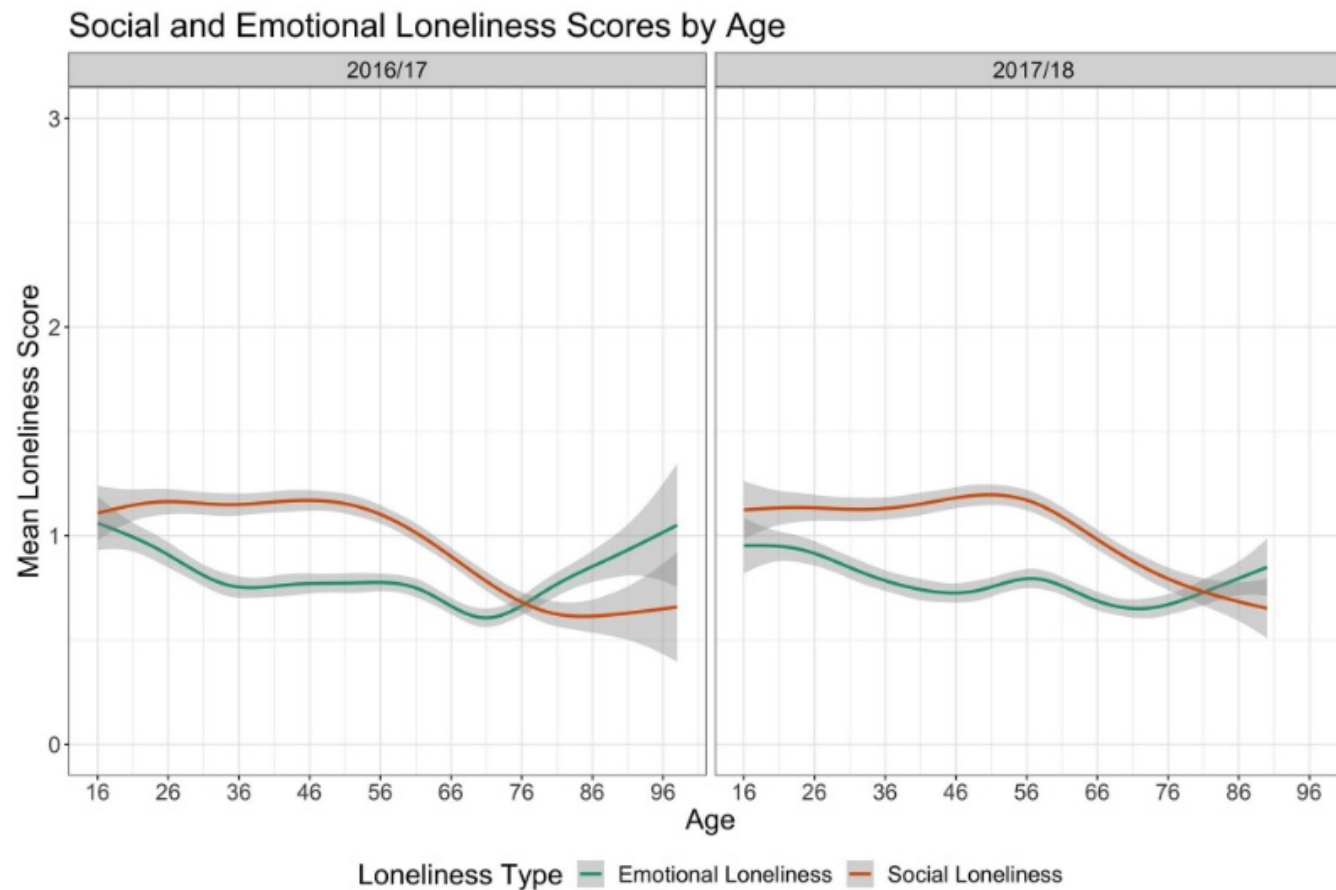
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(Burns et al. 2021)

# Lifespan



# social vs emotional



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(Manoli et al 2022)

# Interventions





# Are interventions effective?

- Sometimes, but not always.
- Effect sizes are modest.
- Not a one-size-fits-all solution.
- We lack a clear mechanistic understanding to guide interventions.



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(Larsgaard et al. 2025; Masi et al. 2011; Morrish et al. 2023)

# Take a break



# Part 2 - Read and discuss



# Discussion material

- break into small groups (~ 5 per group)
- discuss aspects of the lecture
- discuss aspects of these articles:

Is loneliness worse than smoking 15 cigarettes a day?:

BBC's 'More or less' radio programme

There is **NOT** an epidemic of loneliness:

Psychology Today

The Conversation



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# Acknowledgements

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